



INDEPENDENT BA PORTFOLIO CASE STUDY · FITNESS STUDIO ERP · MAY 2026

KAGE Fitness Studio ERP

ERP Overview - Business Analyst Portfolio Case Study

This document summarises the business context, MVP scope, ERP modules, key business logic and validation snapshot for the KAGE Fitness Studio ERP portfolio project.

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Project Type Independent Business Analyst Portfolio Case Study	Domain Fitness Studio / ERP / Gym Operations

Portfolio Notice
All names, figures, workflows and screen designs in this document are fictional and created for portfolio demonstration only. This is not a live client project, production ERP system or final technical implementation.

Project Overview

KAGE Fitness Studio is positioned as a mid-sized gym and wellness centre that runs day-to-day operations through spreadsheets, WhatsApp follow-ups and manual registers. As operations grow, this creates missed renewals, unclear PT session records, informal freelancer billing and delayed management reports.

This BA case study explores how an ERP-style operating model could connect memberships, PT sessions, finance, freelancer access, equipment maintenance and reporting into one structured workflow.

Core ERP Modules	7 modules
UAT Test Cases	26 simulated UAT cases
Process Flows	5 workflow artefacts
Prototype	Clickable front-end demo with role-based views

The objective is not to claim a production build, but to show how a Business Analyst can structure scope, business rules, data relationships, UAT logic and stakeholder-facing prototype communication.

MVP Scope vs Future Scope

The MVP is intentionally focused on operational workflows that matter to a fitness studio owner, front desk team, trainers and finance/admin users.

- Membership creation, tracking and status management - PT session booking, trainer allocation and session logging - Finance module: invoices, payments and due amounts - Freelancer billing and access-fee payment tracking - Renewal reminders and follow-up workflows - Equipment maintenance logs and replacement review scheduling - Reporting dashboard for management - UAT planning, test cases and defect tracking	- Payroll automation for staff - Biometric attendance integration - Payment gateway API integration - WhatsApp Business API integration - GST filing and bank reconciliation - Advanced native mobile app - AI-based predictions or recommendations

Scope clarification
The trainer dashboard is a limited role-based view inside PT Session & Trainer Management. It is not counted as a separate ERP module or full mobile app.

Key Business Problems Identified

1	Scattered membership records	Hard to verify active members; missed renewals and access disputes
2	PT package tracking gaps	Sessions billed incorrectly; remaining sessions unclear
3	Trainer availability not visible	Double bookings and informal client reassignment
4	Renewals depended on memory	Members lapse without consistent follow-up
5	Freelancer billing confusion	Access-fee and client-count calculations disputed
6	Equipment repair history informal	Repeated repairs not visible for replacement review
7	Delayed management reporting	Decisions made with outdated data

AS-IS vs TO-BE: How Things Change

Memberships	Paper/spreadsheet checks	Digital profiles, status updates and expiry visibility
PT Sessions	Trainer notebook and informal counts	Session log, remaining balance and completion control
Trainer Availability	Verbal/WhatsApp checks	Availability view and booking conflict checks
Renewals	Follow-up depends on memory	7-day Renewal Due queue and front-desk follow-up
Freelancer Billing	Manual client/access-fee tracking	Own Fee + active client count x access fee calculation
Equipment Maintenance	Informal repair notes	Issue log, maintenance history and review schedule
Management Reports	Manual spreadsheet compilation	Dashboard for revenue, PT, membership and renewals

Seven Core Modules

Membership Management	Create member profiles, plan linkage, expiry dates, duplicate validation and status visibility.
PT Session & Trainer Management	Allocate trainers, schedule sessions, track attendance, completed sessions and package balance.
Finance & Billing	Track payments, payment mode, invoices, partial/pending amounts and revenue categories.
Freelancer Management	Track freelancer own fee, active client count, monthly payable and access flag status.
Renewals & Retention	Show renewal due, overdue renewal queue and follow-up ownership.
Equipment Maintenance	Log repairs, vendors, repair cost, equipment status and replacement review flag.
Reports & Analytics	Summarise membership, PT, freelancer revenue, maintenance cost and net revenue.

Rules That Make the ERP Believable

Freelancer Billing	$MonthlyPayable = OwnFee + (ActiveClientCount \times PerClientAccessFee)$	Billing is access-fee based, not session-based. Private coaching fee is outside KAGE billing scope.
Trainer Allocation	Availability, current PT load, preferred time slot, client goal and specialisation are considered.	Same trainer should continue where possible, but scheduling can vary by availability.
PT Package Balance	Balance reduces only after SessionStatus = Completed.	Cancelled sessions before start do not reduce package balance.
Commission	CommissionStatus remains Estimated until Admin/Finance approval.	Trainer can view own estimated/approved commission only.
Equipment Review	3 or more repairs in a rolling 6-month window triggers ReplacementReviewFlag.	System flags review; owner/admin decides replacement.
Revenue Report	RevenueReport is a reporting view, not a core transaction table.	It summarises operational data rather than storing every transaction.

UAT Summary

Total Test Cases 26	Passed 20	Failed 4	Pending 2	Initial Pass Rate 77%
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The simulated UAT cycle validates membership, PT sessions, finance, freelancer billing, renewals, equipment and reports. The 77% initial pass rate is intentionally realistic: it shows that defects were identified, prioritised and handled rather than presenting an artificial 100% pass rate.

Critical defects	Actioned before sign-off.
Minor issues	Documented for review or future iteration.
Notification/API items	Deferred because WhatsApp/SMS and external APIs are outside MVP.
Sign-off readiness	Ready as a portfolio validation artefact, not client-signed UAT.

Key Assumptions

- The project uses fictional gym data and simulated user roles.
- Dates and financial values are illustrative demo values.
- Final thresholds such as gate flag days and replacement review period would be validated with business stakeholders in a real project.
- Integrations such as payment gateway, WhatsApp API, biometric attendance and GST filing are excluded from MVP.